Logistic Regression

Lab (8)

1. Breast Cancer Diagnosis (sklearn)

- **Dataset: load_breast_cancer()**
- **Tasks:**
  - Load data and perform EDA (check class balance, feature distributions).
  - Split data (80% train, 20% test).
  - Train logistic regression without regularization (penalty=None).
  - Evaluate using confusion matrix, precision, recall, and ROC-AUC.
  - Add L2 regularization (tune C via grid search).
  - Identify top 3 most important features using coefficients.

2. Iris Flower Binary Classification (sklearn)

- **Dataset: load_iris() (use *setosa* vs. *versicolor* only)**
- **Tasks:**
  - Filter data to two classes and select sepal length and petal width.
  - Plot decision boundary using plt.scatter() and model coefficients.
  - Train model and calculate accuracy.
  - Introduce a dummy feature (e.g., random noise) and observe impact on coefficients.

3. Wine Type Classification (sklearn)

- **Dataset: load_wine() (binary: class_0 vs class_1)**
- **Tasks:**
  - StandardScaler preprocessing.
  - Train model with solver='sag' and compare convergence to 'lbfgs'.
  - Use cross_val_score (k=5) to estimate robustness.
  - Visualize feature importance via horizontal bar chart.

4. Titanic Survival Prediction (Kaggle)

- **Dataset:** Titanic
- **Tasks:**

  - Handle missing Age (impute median) and Embarked (impute mode).
  - One-hot encode Sex and Embarked.
  - Train model using Pclass, Sex, Age, SibSp.
  - Evaluate with ROC curve and calculate Youden's J statistic.
  - Interpret odds ratios for Sex_male and Pclass.

5. Diabetes Prediction (Kaggle)

- **Dataset:** Pima Indians Diabetes
- **Tasks:**

  - Replace Glucose, BMI=0 values with mean.
  - Use LogisticRegressionCV for automated L1 regularization tuning.
  - Plot regularization path for Insulin vs Glucose.
  - Evaluate precision-recall tradeoff using PrecisionRecallDisplay.

6. Spam Email Detection (Kaggle)

- **Dataset:** SMS Spam Collection
- **Tasks:**

  - Preprocess text: lowercase, remove punctuation.
  - Create TF-IDF features (TfidfVectorizer, max_features=500).
  - Train model and identify top 10 spam-indicative words.
  - Test model on custom input: "Congrats! You won $1000.".

7. Credit Card Fraud (Kaggle)

- **Dataset:** Credit Card Fraud Detection
- **Tasks:**

  - Use StandardScaler on Amount and Time.
  - Train model with class_weight='balanced'.
  - Optimize threshold to maximize F1-score.

- Compare precision-recall curves with/without class weighting.

8. Employee Attrition (Kaggle)

- **Dataset:** IBM HR Analytics
- **Tasks:**

  - One-hot encode Department, JobRole, OverTime.
  - Analyze coefficient for OverTime_Yes and MonthlyIncome.
  - Calculate log-loss using log_loss(y_test, y_pred_proba).
  - Deploy model via pickle and predict attrition for synthetic input.

9. Heart Disease Prediction (Kaggle)

- **Dataset:** UCI Heart Disease
- **Tasks:**

  - Handle categorical features: cp, restecg (one-hot encode).
  - Use PolynomialFeatures(degree=2) + feature selection (SelectKBest).
  - Tune tol (tolerance) to control training time.
  - Build calibration curve with CalibrationDisplay.

10. Loan Default Prediction (Kaggle)

- **Dataset:** Loan Data
- **Tasks:**

  - Impute missing Credit_History (mode).
  - Convert Property_Area (rural/semiurban/urban) to ordinal.
  - Train model with stochastic gradient descent solver (solver='saga').
  - Generate SHAP values for explainability.